

## Task 14: Stored Procedures & User-Defined Functions

### Tools:

- MySQL Workbench
- Alternatives: PostgreSQL, BigQuery Sandbox, SQLite

### Hints / Mini Guide:

1. Create a stored procedure to insert employee data.
2. Pass parameters into procedures.
3. Call procedures and verify outputs.
4. Create a user-defined function to calculate tax or bonus.
5. Compare functions vs procedures.
6. Handle errors inside procedures.
7. Optimize reusable business logic.
8. Map procedures to backend services.

### Deliverables:

- SQL file containing procedures and functions

### Final Outcome:

- Intern builds reusable database logic like real backend systems.

### Interview Questions Related To Above Task:

- Difference between procedure and function?
- Why use stored procedures?
- Can procedures return values?
- Security benefits of procedures?
- Where are procedures used in real systems?

## 📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best  
of  
Luck

